

Lecture - 7 Kernels

SVM

- Optimization problem
- Representer theorems
- Kernels
- Examples of Kernels

Recap:

Optimal margin classifier

$$\min_{w, b} \frac{1}{2} \|w\|^2$$

st. $y^{(i)} (w^T x^{(i)} + b) \geq 1$

Assuming $w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$

$$\min_{w, b} \frac{1}{2} \|w\|^2$$

such that $y^{(i)} (w^T x^{(i)} + b) \geq 1 \quad i=1, 2, \dots, m$

Also, $\|w\|^2 = w \cdot w^T$

$$= \min \frac{1}{2} \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T \left(\sum_{j=1}^m \alpha_j y^{(j)} x^{(j)} \right)$$

$$= \min \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y^{(i)} y^{(j)} \underbrace{x^{(i)T} x^{(j)}}_{\langle x^{(i)}, x^{(j)} \rangle}$$

$\langle X, Z \rangle = X^T Z_0$ is the inner product.

Also, $y^{(i)} (w^T x^{(i)} + b) \geq 1$ becomes

$$y^{(i)} \left(\sum_j \alpha_j y^{(j)} x^{(j)} \right)^T x^{(i)} + b$$

$$y^{(i)} \left(\sum_j \alpha_j y^{(j)} \langle x^{(j)}, x^{(i)} \rangle + b \right) \geq 1$$

$$\max \sum \alpha_i - \frac{1}{2} \sum_i \sum_j y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle$$

such that $\alpha_i \geq 0$

" Dual
 optimization
 problem"

$$\sum_i y^{(i)} \alpha_i = 0$$

- 1) solve for α_i 's & b
- 2) to make prediction

$$h_{w,b}(x) = g(w^T x + b)$$

$$= g \left(\left(\sum_i \alpha_i y^{(i)} x^{(i)} \right)^T x + b \right)$$

$$= g \left(\sum_i \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b \right)$$

Kernel Trick

Write algorithm in terms of $\langle x^{(i)}, x^{(j)} \rangle$ (or $\langle X, Z \rangle$)

2) Let there be mapping from $X \rightarrow \phi(X)$
 $1D/2D$

3) Find a way to compute $K(x, z) = \phi(x)^T \phi(z)$

4) Replace $\langle x, z \rangle$ in algorithm 3 with $K(x, z)$

Taking an example $\Rightarrow$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$X \in \mathbb{R}^n$

$$\phi(x) = \begin{bmatrix} x_1 x_1 \\ x_1 x_2 \\ x_1 x_3 \\ x_2 x_1 \\ x_2 x_2 \\ x_2 x_3 \\ x_3 x_1 \\ x_3 x_2 \\ x_3 x_3 \end{bmatrix}$$

$$\phi(z) = \begin{bmatrix} z_1 z_1 \\ z_1 z_2 \\ z_1 z_3 \\ z_2 z_1 \\ z_2 z_2 \\ z_2 z_3 \\ z_3 z_1 \\ z_3 z_2 \\ z_3 z_3 \end{bmatrix}$$

$\hookrightarrow n^2$ elements,
 Need $O(n^2)$ time to
 compute $\phi(x)$ on
 $\phi(x)^T \phi(z)$
 explicitly.

Also, $K(x, z) = \phi(x)^T \phi(z) = (X^T Z)^2$

$\begin{matrix} \uparrow & \searrow \\ \mathbb{R}^n & \mathbb{R}^n \end{matrix}$

$$\begin{aligned}
 &= \left(\sum_{i=1}^n x_i z_i \right) \left(\sum_{j=1}^n x_j z_j \right) \\
 &= \sum_{i=1}^n \sum_{j=1}^n x_i z_i x_j z_j \\
 &= \sum_{i=1}^n \sum_{j=1}^n x_i x_j (z_i z_j)
 \end{aligned}$$

~~$(x^T z)$~~ hence proved.
 $\phi(x)^T \phi(z)$

SVM = optimal margin classifier + Kernel Trick.

How to make kernels?

If x, z are "similar" $K(x, z) = \phi(x)^T \phi(z)$ is "large"

If x, z are "dissimilar" $K(x, z)$ is "small"

$$K(x, z) = \exp \left(- \frac{\|x - z\|^2}{2\sigma^2} \right)$$

$\downarrow$
 does exist ϕ such that $K(x, z) = \phi(x)^T \phi(z)$
 $K(x, x) = \phi(x)^T \phi(x) \geq 0 \rightarrow$ A valid
 Kernel function.

Let $\{x^{(1)}, \dots, x^{(d)}\}$ be d points

Let $K \in \mathbb{R}^{d \times d}$ "kernel matrix"

$$K_{ij} = K(x^{(i)}, x^{(j)})$$

Given any vector z ,

$$z^T K z = \sum_i \sum_j z_i K_{ij} z_j$$

$$= \sum_i \sum_j z_i \phi(x^{(i)})^T \phi(x^{(j)}) z_j$$

$$= \sum_i z_i \underbrace{\sum_j \phi(x^{(j)})^T \phi(x^{(i)})}_{K} z_j$$

$$\sum_k \underbrace{\left(\sum_i z_i \phi(x^{(i)})^T \right)}_K \underbrace{\phi(x^{(k)})}_{K} z_k$$

$$\sum_k \left(\sum_i z_i \phi(x^{(i)})^T \right)_K^2 \geq 0, \text{ which}$$

proves that $K \succeq 0$.

It turns out to be the sufficient condition, where then K is a valid kernel function.

Theorem (Mercer): K is a valid kernel function (i.e. $\exists \phi$ such that $K(x, z) = \phi(x)^T \phi(z)$) if and only if for any d points $\{x^{(1)}, x^{(2)}, \dots, x^{(d)}\}$, the corresponding kernel matrix $K \succeq 0$.

$$K(x, z) = \exp\left(-\frac{\|x - z\|^2}{2\sigma^2}\right) \rightarrow \phi(x) \in \mathbb{R}^\infty$$

↓ gaussian kernel

$$\text{Linear Kernel} = K(x, z) = x^T z$$

$$\phi(x) = x$$

Linear kernel is the most widely used, which basically means not using a kernel at all, as we don't go to higher dimensional space.

The second most widely kernel is gaussian kernel.

L1 Norm Soft Margin SVM

$$\min_{w, b, \xi_i} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i$$

such that

$$y^{(i)} (w^T x^{(i)} + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0$$

After applying the representer theorem,

$$\max_{\alpha} W(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j K(x^{(i)}, x^{(j)})$$

such that $0 \leq \alpha_i \leq C$ $i=1, 2, \dots, m$

$$\sum_{i=1}^m \alpha_i y(i) = 0$$

This is known as dual form.

$$K(x, z) = (x^T z)^d \rightarrow \text{polynomial kernel}$$

$$K(x, z) = \exp\left(-\frac{\|x - z\|^2}{2\sigma^2}\right) \rightarrow \text{gaussian kernel}$$

$$\frac{1}{2} \|w\|^2 + \sum_{i=1}^n y_i (w^T x_i - 1)$$

$$-1 \leq (1 + w^T x_i) y_i$$

$$0 \leq 1 + w^T x_i y_i$$

$$\sum_{i=1}^n \frac{1}{2} = \frac{1}{2} n$$